How accurate LIGHT v0_2 is

The basic network of LIGHT v0_2, version 0.1.2, on 30 September 2026, with its energy ladder, its system map and its reaction rates

Bottom line. LIGHT v0_2 reproduces the fitted KrF decays [1, 2] within 2.3 to 6.1 % rms and predicts decays that no fit used [4] within 11.7 % rms, and at the Tahoma laser [5] its default network gives 89 % of the measured small-signal gain of 6.39 %/cm.

It also gives 111 % of Tahoma's saturation intensity and 99 % of its formation efficiency. The C/B emission ratio of Yu and Setser [3] is matched to -0.5 %. No fit used Tahoma's numbers and no rate was set by a laser measurement, so the Tahoma scores are predictions. Most of the gain it misses lies inside the band of Kr_2^* + F → KrF(B) + Kr, a rate no one has measured, whose top gives 97 %.

Its largest misses are on data that conflict with the fitted sets, 19.3 and 34.1 % rms on the C-A decays of Hakuta et al. [6] and -22.3 % on the KrF(B) lifetime of Burnham and Searles [7], and the VUV yield of pure krypton at low pump, 136 % of Morozov's measurement [8].

[1] Eden, Waynant, Searles and Burnham, Appl. Phys. Lett. 32, 733 (1978). [2] Gadomski, Xu, Setser and Nelson, Chem. Phys. Lett. 189, 153 (1992). [3] Yu and Setser, J. Phys. Chem. 94, 2934 (1990). [4] Xu, Gadomski and Setser, J. Chem. Phys. 99, 2591 (1993). [5] Spectra Technology, *Experimental Support for KrF Laser Modeling*, final report to Los Alamos, 19 July 1989, OSTI 10136715. [6] Hakuta et al., Jpn. J. Appl. Phys. 29, 1194 (1990). [7] Burnham and Searles, J. Chem. Phys. 67, 5967 (1977). [8] Morozov, Heindl, Krücken, Ulrich and Wieser, J. Appl. Phys. 103, 103301 (2008).

LIGHT v0_2 is v0_1's working copy. Its input is v0_1's, a power deposited uniformly in the electrons, and its basic network adds the Kr_2^* route to v0_1's, 44 reaction rows in all. Every rate known only as a published spread, or not at all, is carried as a band. A band's default is a measurement, or 0 where no measurement exists, so the default network rests on measurements and theory alone. The KrF(B,C) block and every formation link are v0_1's. Every KrF rate in the block but one is a measurement of that process or a joint fit to kinetic measurements of it, and the exception is the KrF(B) radiative lifetime, Kannari's 6.99 ns where Eden measured 6.8 ± 0.2 ns. No rate was set by a laser measurement. Scored against Tahoma, the one independent Kr/F_2 laser, under the same beam drive as v0 and v0_1, v0_2's default network gives the small-signal gain at $\times 0.89$ the measured value, the saturation intensity at $\times 1.11$ and the formation efficiency $g_0 I_{sat}/P_{in}$ at $\times 0.99$. Kannari's estimates for the two unmeasured Kr_2^* rows give the gain at $\times 0.95$, and Kr_2^* + F at the top of its band gives $\times 0.97$. The largest open questions are the rate of Kr_2^* + F → KrF(B) + Kr, which has no measurement, and the trimer rate above 425 K. Table 1 lists every test. The comparison with v0 and v0_1 is in `docs/REVIEW/`.

1 How the numbers were made

Each dataset has one role. Kinetic measurements of the channels being fitted train them. Other kinetic measurements are held out and test the fit. Laser performance only scores, and the fitting code refuses a dataset of that kind. This is the repository's predictive-only policy. Measurements score the model after the fact and never steer a number, and a band is never narrowed or moved by a score. Every number in this note is written by `docs/make_overview_v0_2.py` and `docs/make_system_map_v0_2.py` from v0_2 and from `ML/`, and a test fails when those files no longer match the model.

ML runs v0_2 when the environment variable `LIGHT_VERSION` names it. v0_2 takes only a power input. Tahoma was pumped by an electron beam, so v0_2 is scored there under the same prototype beam drive as v0 and v0_1, a test fixture in `docs/` and not part of the model. Since v0_2's default network carries the Kr_2^* route, ML's Tahoma configuration is that default network, with every band at its default and n_e free from a seed of 1.0×10^{10} cm^{-3}. Section 8 runs v0_2's own input at Tahoma's conditions. Section 5 moves one band at a time to each end of its range, the others at their defaults.

test	measured	v0_2	role of the data
energy ledger closure, 9 runs across the modes and bands		below 10^{-12} of E_{dep}	bookkeeping
KrF(B) decay against Kr and F_2, Eden 1978		rms 2.3 and 3.2 %	train
KrF(B,C) slow decay, Gadomski 1992 Figs. 1 and 2		rms 6.1 and 4.2 %	train
C/B emission ratio at 300 K, Yu and Setser 1990	0.16 ± 0.02	-0.5 %	train
KrF(B,C) decays, Xu 1993 Figs. 5 to 7		rms 11.7 %	held out
decay constants Xu 1993 printed, 50 and 100 Torr Kr	5.0×10^{7} and 6.2×10^{7} s^{-1}	$\times0.91$ and $\times0.97$	structure check
C-A decays, Hakuta 1990		rms 19.3 and 34.1 %	held out, in conflict
τ_B, Burnham and Searles 1977	9.0 ± 0.6 ns	-22.3 %	held out, in conflict
C/B ratio 230 K over 300 K, Yu and Setser	1.30 ± 0.06	-8.4 %	held out, partly behind the gap v0_2 books
Kr^* + Kr^+ made per eV, pure Kr and Ar, Saito 2003		$\times1.02$ and $\times1.03$	check
Kr_2^+ + e^- recombination, Shiu and Biondi 1977		$\times1.00$	v0_2's input
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$, Smith et al. 1972	2.74×10^{-31} to 3.3×10^{-31}	2.4×10^{-31}	in v0_2's input
VUV yield of pure Kr at low pump, Morozov 2008	0.42 ± 0.05	0.57, $\times1.36$	check
the same, each band at either end		moves by 0.001 or less	check
electron closure at Tahoma, kinetic EEDF loop	T_e 1.61 eV	1.60 eV	reference
Tahoma small-signal gain g_0, v0_2's default network	6.39 %/cm	5.69, $\times0.89$	device, score only
the same, Kannari's two Kr_2^* estimates		6.10, $\times0.95$	device, score only
the same, Kr_2^* + F at the top of its band		6.23, $\times0.97$	device, score only
the same, trimer on $T_g^{-1/2}$		2.84, $\times0.44$	device, score only
Tahoma saturation intensity I_{sat}	1.4 ± 0.1 MW/cm^2	1.55, $\times1.11$	device, score only
Tahoma formation efficiency $g_0 I_{sat}/P_{in}$	0.27	0.268, $\times0.99$	device, score only

Table 1: LIGHT v0_2 against the data, every band at its default unless a row says otherwise. Rates are in cm^{3}s^{-1} or cm^{6}s^{-1}. A percentage is v0_2's deviation from the data, signed for a single value and rms for a set. A $\times$ factor is v0_2 over the measurement.

2 The KrF(B,C) block

v0_2 keeps v0's KrF(B,C) block, as v0_1 does. KrF has two emitting states. B lases at 248 nm, and C radiates slowly in a band at 275 nm. LIGHT treats them as the two coupled levels of eq. 5 of Gadomski et al. (1992) and reproduces that model's analytic decay rate to within 1.7×10^{-4} at every pressure tested. The block has 10 constants. Nine come from one joint fit to three measurement sets that agree with one another, the KrF(B) decays of Eden et al. (1978), the Kansas State B/C decays (Gadomski et al. 1992) and Yu and Setser's equilibrium C/B ratio (1990). The tenth, τ_B, stays at Kannari's 6.99 ns, inside the fit's interval of 6.79 to 7.22 ns. v0_2 shares it between the 248-nm B-X band and B-A emission at 272 nm, which by Dunning and Hay's calculation carries 3.5 % of the B-X rate, so the total radiative rate of B is unchanged. The fit's median and 68 % interval for B-C mixing by Kr is k_{BC} = 9.8×10^{-12} [4.63×10^{-12}, 1.73×10^{-11}] cm^{3}s^{-1}. The equilibrium ratio is K_{eq} = 1.07 [0.84, 1.38], and the C lifetime is τ_C = 50 [41, 63] ns. LIGHT carries the best fit, k_{BC} = 1.05×10^{-11} cm^{3}s^{-1}, K_{eq} = 1.10 and τ_C = 50 ns. At 1 atm and 300 K the lasing B state holds 0.69 [0.57, 0.80] of the manifold, and LIGHT's rounded constants give 0.68.

The fitted decay sets are matched to between 2.3 and 6.1 % rms. The held-out decays of Xu et al. (1993) are matched to 11.7 %, and the two decay constants they print come out at $\times0.91$ and $\times0.97$. Hakuta et al. (1990) measured faster C-A decays, missed by 19.3 and 34.1 % rms, and their data conflict with the three fitted sets. Burnham and Searles (1977) measured τ_B = 9.0 ns. Those two conflicts are v0's and remain.

The energy ledger books Yu and Setser's measured B-C gap, 120 cm^{-1}, as v0_1 does. It sets how K_{eq} changes with temperature, and their C/B ratio between 230 and 300 K comes out -8.4 %. That is not an independent test. Their 120 ± 50 cm^{-1} is a consensus of two routes, 110 ± 20 from the absolute ratio and 180 ± 30 from this very pair of temperatures.

Tahoma's gain is the sharpest test of the block, since g_0 is the KrF(B) density times the stimulated cross section. Over 40 draws from the fit's posterior the gain is 5.39 %/cm [4.99, 6.18] against the measured 6.39.

3 Charge, recombination and fluorine

The charge chemistry is v0_1's, and no electrons are injected. n_e starts from a seed of 10^{15} cm^{-3} and follows ionization, attachment and recombination. At the reference point it runs at 1.68×10^{14} cm^{-3} while F_2 lasts and peaks at 1.70×10^{15} cm^{-3} after F_2 burns out at 640 ns. In three-body collisions with Kr, Kr^+ turns into Kr_2^+ at 2.4×10^{-31} cm^6s^{-1}, the mean of three measurements. Kr_2^+ recombines with electrons at the rate Shiu and Biondi measured, $1.6\times10^{-6}\,(300\,K/T_e)^{0.55}$ cm^3s^{-1}. Each recombination returns a Kr^* and heats the gas. Of the ions that recombine, 88.7 % meet an electron and 11.3 % meet F^- and form KrF(B), which makes 4.7 % of the KrF formed. Once the F_2 is gone, Penning ionization of Kr^* and recombination hand the same energy back and forth, so the flow through Kr^* passes 100 % of E_{dep}, 113.8 % here. In v0_2 part of that Kr^* forms Kr_2^* and radiates at 147 nm instead (Section 4), and the gas ends at 649 K.

F + F + Kr gives F atoms back to F_2 at 2.8×10^{-34} cm^6s^{-1}, Cohen and Bott's evaluated Ar value, since no Kr measurement exists. At the reference point it returns 0.1 % of E_{dep} from the F atoms to heat, far too slowly to refill the F_2.

4 The Kr_2^* route

v0_2's basic network carries the Kr_2^* excimer, which v0_1 had as an optional group. $Kr^* + 2Kr \rightarrow Kr_2^* + Kr$ runs at Ueda's 1.7×10^{-32} cm^6s^{-1}, measured at 0.5 to 1 atm, on Wieme and Lenaerts' $(T_g/300)^{-0.64}$, measured over 300 to 451 K. Kr_2^* radiates at 147 nm with the measured 270-ns lifetime. $Kr_2^* + F_2$ runs at Quigley and Hughes' 2.1×10^{-10} cm^3s^{-1}, split evenly between KrF(B) and Kr_2F^* by Kannari's assignment, since the products were not measured. $Kr_2^* + F \rightarrow$ KrF(B) + Kr and $Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$ have no measurement and run at 0. At the reference point 25.6 % of the Kr^* forms Kr_2^*, 15.1 % of the Kr^* used while the F_2 lasts. 80.8 % of that Kr_2^* radiates, so 20.0 % of E_{dep} leaves at 147 nm, and $Kr_2^* + F_2$ forms 5.4 % of the KrF.

$Kr_2^* + F$ decides how much of the route reaches the laser line. With no measurement its band runs from 0 to 1.0×10^{-9} cm^3s^{-1}, the top of a conditional range that SRI gave for the argon analog. At the reference point the 248-nm share rises from 4.67 % at 0 to 6.52 % at 1.0×10^{-11}, 9.21 % at Kannari's 3.0×10^{-10} and 9.50 % at the top, where the 147-nm share falls from 20.0 to 0.5 %. The share saturates because the F_2 burns out at 640 ns and leaves 2.12×10^{17} cm^{-3} F atoms, which turn Kr_2^* into KrF 17 times faster than Kr_2^* radiates at Kannari's rate and 57 times faster at the top, and each KrF photon gives its F atom back. At the top 74 % of the gain comes after the burnout. With Kannari's estimates for both unmeasured rows, 3.0×10^{-10} and 5.0×10^{-10} cm^3s^{-1}, v0_2 gives v0_1's optional Kr_2^* group, 9.17 %.

Pure krypton tests the route without F_2. At Morozov's low pump v0_2 gives a VUV yield of 0.57, where Morozov et al. (2008) measured 0.42 ± 0.05, and no band at either end moves it by more than 0.001. So the miss does not come from the route, and that point cannot score the bands. At 300 kW cm^{-3} for 1 μs in 665 Torr of Kr the bands move the yield from 0.13, with $Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$ at 5.0×10^{-10} $cm^3\,s^{-1}$, through 0.27 at the defaults to 0.49, with $Kr^* + 2Kr \rightarrow Kr_2^* + Kr$ at 5.5×10^{-32} $cm^6\,s^{-1}$. A pure-Kr VUV yield measured against pump rate would score them, and none is among the sources held.

5 The bands

v0_2 carries 8 rates as bands. A measured rate keeps its measurement as the default and spans the other published values. A rate with no measurement is 0 by default and runs to the largest published value, which is no bound, and 3 bands are of that kind. Analogs and device data never set a default or narrow a band. Appendix A.4 gives each band's sources. Table 2 moves one band at a time to each end at the reference point. Two bands move the 248-nm share far, Kr_2^* + F from 4.67 to 9.50 % and the trimer's temperature law from 4.67 to 2.22 %. Each of the other six moves it by 0.21 points or less.

band	default	low to high end	248 nm	147 nm	T_g at the end (K)
Kr_2^* + F → KrF(B) + Kr	0	0 to 1.0×10^{-9} cm^3 s^{-1}	4.67 to 9.50	20.0 to 0.5	649 to 726
the trimer's $(T_g/300)^n$	−3.32	−3.32 to −0.50	4.67 to 2.22	20.0 to 19.5	649 to 658
Kr^* + Kr^* → Kr^+ + Kr + e^-	5.0×10^{-10}	1.5×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	4.87 to 4.67	25.0 to 20.0	616 to 649
Kr_2F^* + F_2	1.2×10^{-10}	1.2×10^{-10} to 4.3×10^{-10} cm^3 s^{-1}	4.67 to 4.88	20.0 to 19.9	649 to 663
Kr_2F^* + F → 2Kr + 2F	0	0 to 1.8×10^{-10} cm^3 s^{-1}	4.67 to 4.87	20.0 to 19.9	649 to 670
Kr^* + 2Kr → Kr_2^* + Kr, 300 K	1.7×10^{-32}	1.7×10^{-32} to 5.5×10^{-32} cm^6 s^{-1}	4.67 to 4.52	20.0 to 28.0	649 to 601
Kr_2^* + F_2, both channels	2.1×10^{-10}	2.1×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	4.67 to 4.60	20.0 to 19.1	649 to 654
Kr_2^* + Kr_2^* → Kr_2^+ + 2Kr + e^-	0	0 to 5.0×10^{-10} cm^3 s^{-1}	4.67 to 4.65	20.0 to 10.8	649 to 704

Table 2: Each band alone at its two ends at the reference point, the others at their defaults. The shares are percent of E_{dep}. With every band at its default v0_2 sends 4.7 % out at 248 nm and 20.0 % at 147 nm, and the gas ends at 649 K.

Tahoma scores the bands and sets none (Table 3). Its gain moves most with the trimer's weak law, to ×0.44, so Tahoma counts against that law. Kr_2^* + F at the top of its band brings the gain to ×0.97. Each of the other six moves it by 0.38 %/cm or less. Tahoma's window, 200 to 400 ns, closes with 29 % of the F_2 fill left, while at the reference point 74 % of the band's gain comes after the F_2 has burnt out. So Kr_2^* + F moves Tahoma's gain far less than the reference point's 248-nm share.

band	g_0 (%/cm), low to high end	over the measured 6.39
Kr_2^* + F → KrF(B) + Kr	5.69 to 6.23	×0.89 to ×0.97
the trimer's $(T_g/300)^n$	5.69 to 2.84	×0.89 to ×0.44
Kr^* + Kr^* → Kr^+ + Kr + e^-	5.71 to 5.69	×0.89 to ×0.89
Kr_2F^* + F_2	5.69 to 5.89	×0.89 to ×0.92
Kr_2F^* + F → 2Kr + 2F	5.69 to 5.86	×0.89 to ×0.92
Kr^* + 2Kr → Kr_2^* + Kr, 300 K	5.69 to 5.31	×0.89 to ×0.83
Kr_2^* + F_2, both channels	5.69 to 5.74	×0.89 to ×0.90
Kr_2^* + Kr_2^* → Kr_2^+ + 2Kr + e^-	5.69 to 5.69	×0.89 to ×0.89

Table 3: Each band alone at its two ends at Tahoma, under the beam drive with ML's constants, the others at their defaults. With every band at its default the gain is 5.69 %/cm.

6 KrF formation

The chain from Kr^+ and Kr^* to KrF is v0_1's. Kr^* + F_2 reacts at 8.47×10^{-10} cm^{3}s^{-1}, the mean of three first-hand measurements. Velazco, Kolts and Setser (1976) measured the KrF^* yield of that reaction as 90 %, and Yu and Setser (1990) put 38 % of that KrF^* in C. Tahoma measured $g_0 I_{sat}/P_{in}$ = 0.27. ML

puts a 248-nm probe flux on v0_2 and fits its I_{sat} at fixed g_0, as Tahoma did. That gives 0.268. Its KrF formation rate in 5-eV quanta is 0.300 of the pump. The difference is KrF that forms in C, or mixes into it, and decays there without meeting the laser field.

Table 4 moves one link at a time to another value it could take. The measured range of the harpoon yield, 0.70 to 1.0, moves $g_0 I_{sat}/P_{in}$ from 0.240 to 0.282 and brackets Tahoma's 0.27. Three links have no measurement at all. They are the KrF^* yield of ion-ion recombination with F^-, the share of that KrF formed in C, and $Kr_2^* + F \rightarrow KrF$. Only analogs exist for the first two, and both analogs lower $g_0 I_{sat}/P_{in}$ toward the measurement. v0_2 runs the third at 0, and Kannari's estimate or the top of its band raise $g_0 I_{sat}/P_{in}$. The predictive-only policy lets no analog set a link and no link be chosen by its fit to Tahoma. The rest of the Kr_2^* route moves $g_0 I_{sat}/P_{in}$ by 0.013 at most.

link	v0_2	tried	what the value is	$g_0 I_{sat}/P_{in}$	change
KrF^* yield of $Kr^* + F_2$	0.90	0.70	the measured 0.90 less its stated 0.20	0.240	−0.028
		1.0	the published assignment	0.282	+0.014
		0.5	as for $Ar^* + F_2$	0.211	−0.057
share of that KrF formed in C	0.38	0	if nascent C relaxes to B at 1 atm	0.290	+0.022
KrF^* yield of ion-ion recombination	1	0.64	as for Cl^-	0.225	−0.043
share of that KrF formed in C	0	0.38	as for SF_6^-	0.251	−0.017
ion-ion recombination rate	Flannery	×0.38	Avco's model input	0.266	−0.002
$Kr_2^* + F \rightarrow KrF(B) + Kr$	0	3.0×10^{-10}	Kannari's estimate	0.285	+0.017
		1.0×10^{-9}	the top of its band	0.290	+0.022

Table 4: Each formation link moved alone from v0_2's chain at Tahoma, 2.8 Torr of F_2 in Kr at 665 Torr in all, pumped at 330 kW cm^{-3}. v0_2's chain gives 0.268, and Tahoma measured 0.27.

7 The electron side

v0_2 integrates each krypton electron-impact rate over v0's EEDF with two temperatures, a Maxwellian bulk at T_e and a tail above 9.915 eV read from kinetic EEDFs. Each captured electron takes the energy it really carries. For F_2 attachment at Tahoma that is 0.55 eV, where $\frac{3}{2}T_e$ would book 2.41 eV. Kr_2^+ recombination goes as $T_e^{-0.55}$, so a recombining electron takes 0.95 T_e. A kinetic loop that iterates v0_2 with a Boltzmann solution, with the same recombination law as its sink, is the reference. At Tahoma v0_2's own closure lands within 0.7 % of the loop's T_e and 2.5 % of its n_e, and within 1.0 and 0.5 % of its gain and formation.

The cross sections are v0's. With Biagi's krypton set the computed W is 0.96 to 1.03 times Nakamura's measured values. For F_2 attachment the groups that measured the rate constant disagree beyond their errors, and SIGLO gives between 0.48 and 2.41 times their values.

On pure gases the beam drive's Spencer-Fano yields make 1.02 times the Kr^* and Kr^+ per eV that the measured W and N_{ex}/N_i give in Kr, and 1.03 times in Ar. v0_2 takes its Kr_2^+ recombination from Shiu and Biondi and its Kr^+ conversion from a mean that includes Smith et al., so two of v0's pure-gas checks are v0_2's inputs. The VUV yield of pure Kr at low pump is still the one miss. Without the thermal electrons' excitation v0_2 gives 0.55, close to the one photon per excitation and per ion that the measured W and N_{ex}/N_i allow, 0.54, so the shortfall is either physics LIGHT lacks or a systematic error of the measurement.

8 The power input

v0_2 feeds its power to the electron gas. Run that way at Tahoma's conditions, with no beam, it gives g_0 = 5.53 %/cm, $\times$0.87 the measured value. Its electrons then carry the power at T_e = 3.8 eV and n_e = 2.59$\times10^{14}$ cm^{-3}, where the beam-sustained plasma sits at 1.6 eV and 6.33$\times10^{14}$ cm^{-3}. The gain comes near, but the electron gas does not resemble the beam's, so the agreement does not test the power input. No discharge-pumped Kr/F_2 measurement has been scored.

9 The energy ladder

Figure 1 places every species of v0_2 at its formation energy, the energy above ground-state Kr and F_2 at rest that Appendix A lists, and draws the rows that move energy between them. It follows the ladder view of the website's excimer simulation, with v0_2's own numbers. KrF(B) lies 5.54 eV below the ion pair $Kr^+ + F^-$, the measured $D_e(B)$, and 5.06 eV above Kr + F. The 248-nm photon carries 5.00 eV of that and leaves KrF(X) only 0.06 eV above Kr + F. The harpoon G3 heats the gas by 3.26 eV for each KrF(B) it makes, the ion-ion recombination K26 by 5.54 eV and the trimer's formation by 0.58 eV. Kr_2F^* emits 2.95 eV at 420 nm and leaves 1.53 eV of lower-state repulsion. On the Kr_2^* route, which is basic in v0_2, Kr_2^* emits 8.43 eV at 147 nm and leaves 0.77 eV, or meets F_2 and makes KrF(B) through NK7 or Kr_2F^* through K2F_BR.

10 The system map

The map on the next page draws v0_2's basic network at the notes' reference point, 1 atm of Kr with 0.5 % F_2, 300 kW cm^{-3} for 1 μs, with n_e seeded at 10^{15} cm^{-3} and then free and every band at its default. Band widths are proportional to the energy they carry, and every box conserves it (43 balances, each to 10^{-8} of E_{dep}). Of the deposited energy, 23.8 % passes through KrF in 5-eV quanta. Of that KrF, 19.6 % radiates at 248 nm, 62.0 % forms the trimer and 5.4 % is quenched by electrons, so 4.7 % of the deposited energy leaves at 248 nm, and 20.0 % leaves at 147 nm from Kr_2^*.

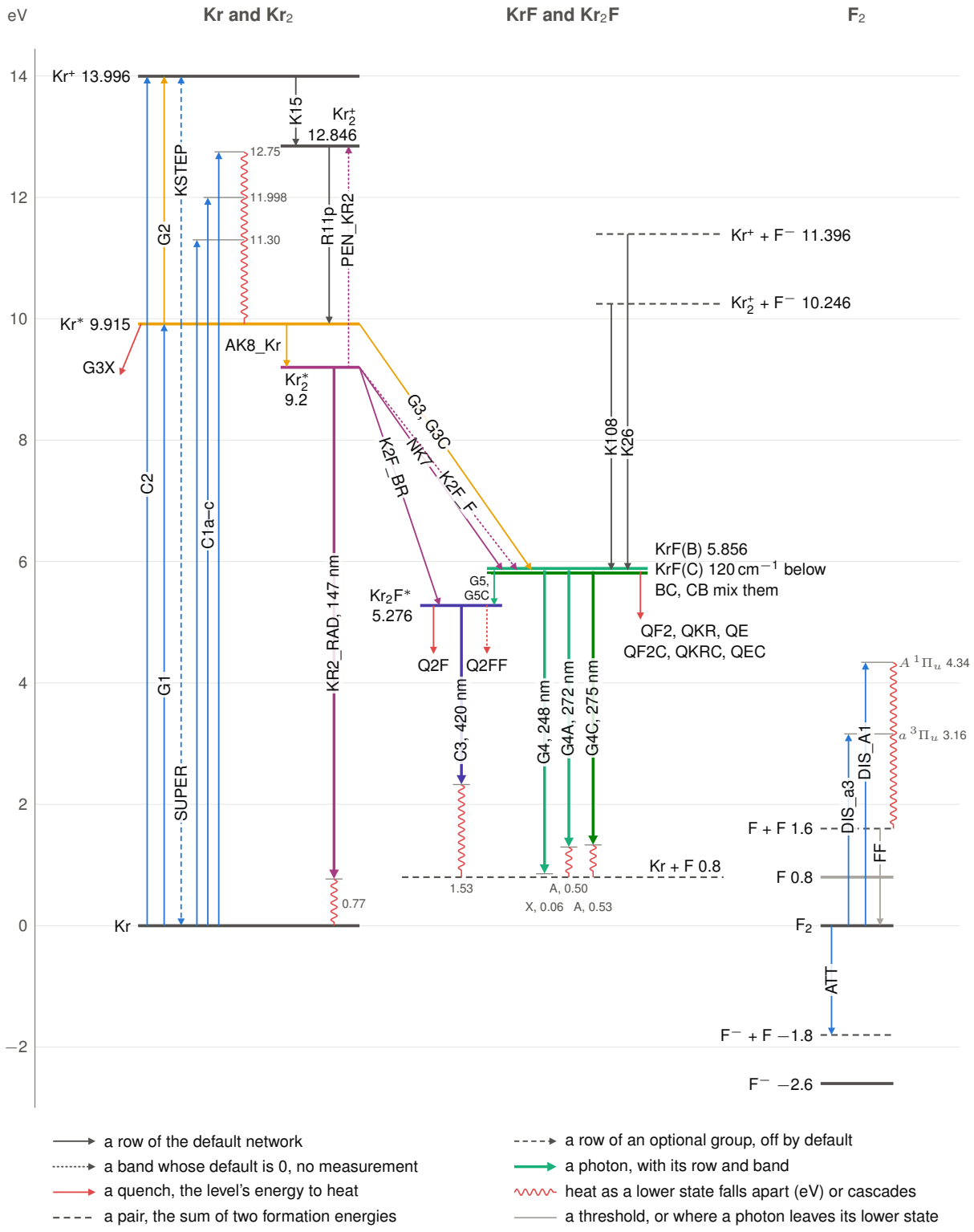
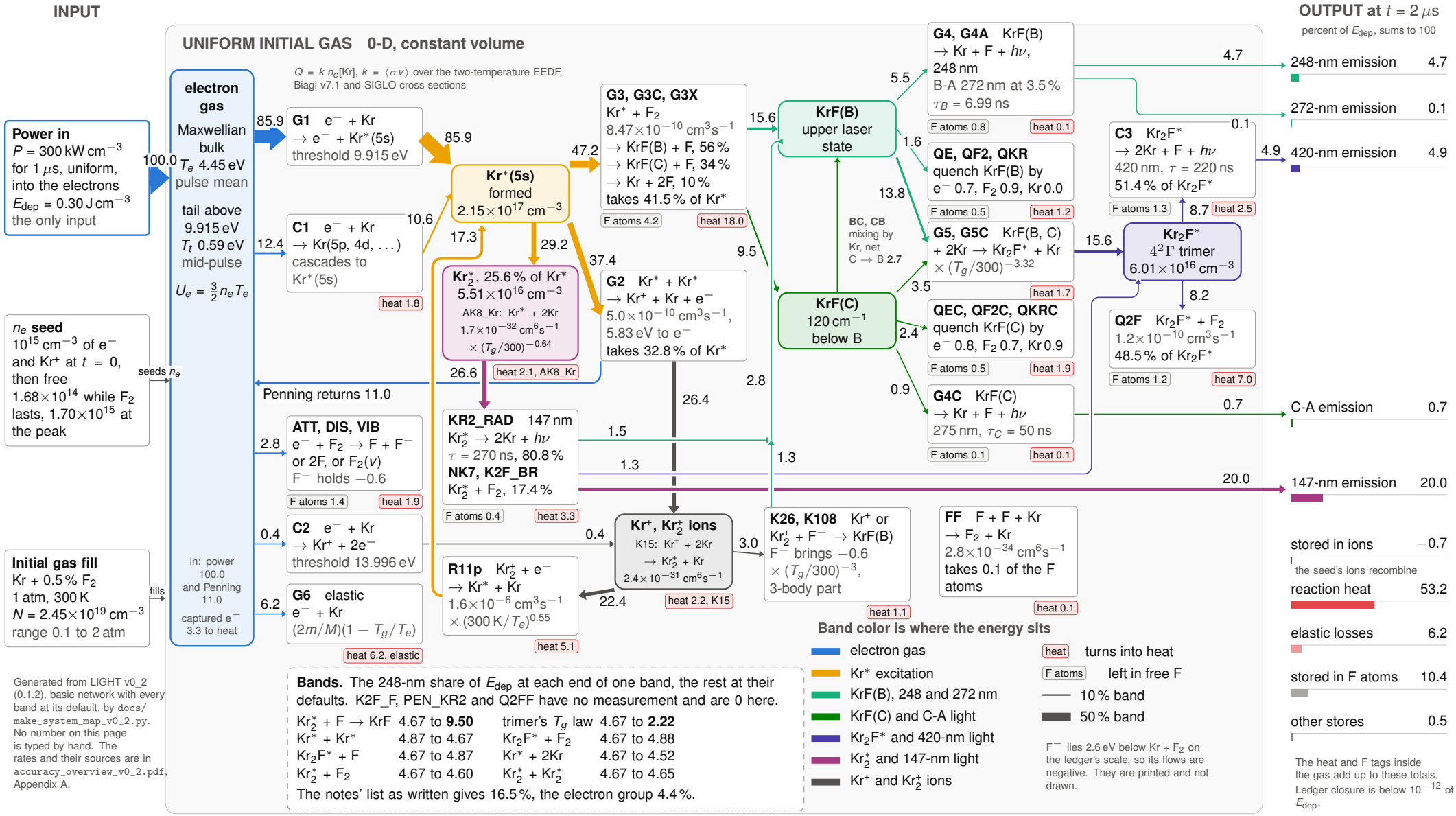


Figure 1: The energy ladder of LIGHT v0_2, version 0.1.2, in eV above ground-state Kr and F_2 at rest. A bar is a species at its formation energy and a dashed bar a pair, the sum of two. An arrow is a row, labelled with its id. It is solid in the default network and dotted for a band whose default is 0, Q2FF, K2F_F and PEN_KR2. The rows of the optional group `electron` are dashed. A photon's arrow drops by its quantum, and the red squiggle below it is the heat its lower state gives the gas as it falls apart. A red stub is a quench. The energy a row releases heats the gas, except a photon's, which leaves, and that of G2, PEN_KR2 and SUPER, which goes to the electrons. KrF(B) and KrF(C) lie 120 cm^{-1} apart and are drawn as a pair of bars that every B and C row meets. The rows G6, VIB1, VIB2, VIB3 and VIB4 move nothing between levels and are not drawn.

LIGHT system map, v0_2: energy flow through the Kr/F_2 kinetics

Reference point: 1 atm of Kr with 0.5 % F_2 at 300 K, P = 300 kW cm^{-3} for 1 μs so that E_{dep} = 0.30 J cm^{-3}, n_e seeded at 10^{15} cm^{-3} and then free, every band at its default, run to 2 μs. Band width is proportional to the energy flow. Numbers are percent of E_{dep}, rounded to 0.1.



The notes' outputs as flows, which overlap: to Kr* 98.3 %, to Kr+ 26.8 %, to KrF(B) and KrF(C) 23.8 %, heat 53.2 %, elastic losses 6.2 %, emission 30.4 %. Of the Kr* formed, 25.6 % forms Kr_2^* (15.1 % of the Kr* used before the F_2 burns out), and 80.8 % of that Kr_2^* radiates at 147 nm. Of the ions that recombine, 88.7 % meet an electron and return to Kr*, and 11.3 % meet F^- and form KrF(B). Of the KrF formed, 8.93×10^{16} cm^{-3}, 4.7 % came from ion-ion recombination and 5.4 % from $Kr_2^* + F_2$. Of it 19.6 % radiates at 248 nm, 0.7 % at 272 nm and 3.3 % in the C-A band, 62.0 % forms the trimer, and electrons, F_2 and Kr quench 5.4 %, 5.7 % and 3.4 %. At the end of the run T_g = 649 K and p = 2.18 atm, and F_2 fell below 10 % of the fill at 640 ns.

11 Where v0_2 is not yet reliable

- Kr_2^* + F → KrF(B) + Kr. No measurement exists, so v0_2 runs it at 0 and carries it as a band up to 1.0×10^{-9} cm^{3}s^{-1}. Kannari's 3.0×10^{-10} traces to an SRI report whose model leaves the reaction out and whose one number is a conditional range for Ar_2^* + F. At the reference point the band spans 4.67 to 9.50 % at 248 nm, and at Tahoma 5.69 to 6.23 %/cm. A measurement or a bound between 10^{-12} and 10^{-10} cm^{3}s^{-1} would matter most.
- The trimer above 425 K. KrF + 2Kr → Kr_2F^* takes 62.0 % of the KrF at the reference point, where the gas ends at 649 K. The $(T_g/300)^{-3.32}$ law is Shui's calculation, measured only up to 425 K. With the pure-Kr $T_g^{-0.5}$ law, the weakest the data allow, the reference point sends 2.22 % out at 248 nm instead of 4.7 %. Tahoma's window sits at 446 K, and there that law gives the gain at ×0.44 the measured value. Tahoma's gain counts against the weak law as a score and sets nothing.
- Kr^* + 2Kr above 451 K. Its temperature law is measured over 300 to 451 K, and the reference point ends at 649 K. The report warns when the gas passes it.
- Hot gas. Recombination returns the ionization energy as heat. At 0.1 atm and 500 kW cm^{-3} the gas reaches 3458 K, beyond every rate law's data, and Kr_2F^* and Kr_2^+ redissociation are not in the model. The report warns above 1000 K.
- Ion-ion recombination with F^-. Its rate was measured only below 300 Torr, where Jacob et al. (1978) fitted KrF* pulse shapes. Rokni et al. (1977) found that Kr_2^+ + F^- forms KrF* as efficiently as Kr^+ + F^-, but from relative intensities. The absolute KrF* yield, its B/C split and the rate near 1 atm have no measurement. Table 4 shows what the analogs would do.
- Kr_2^* + Kr_2^* and Kr^* + Kr^*. Neither Penning rate has a measurement among the sources held, and at high pump they set the pure-Kr VUV yield (Section 4).
- Kr_2^*'s well. REBUILD's estimate puts Kr_2^* 0.715 eV below Kr^* + Kr, since no measured or calculated binding energy is among the sources held. It only moves energy between heat and storage.
- F + F + Kr is held at its 300 K value, which overstates it in hot gas.
- Hakuta's C-A decays and Burnham and Searles' τ_B remain unexplained conflicts.

A Reaction rates

Rates are the values at 300 K. The column *T* gives the law v0_2 applies to them. Electron-impact rows use $\langle\sigma v\rangle$ over the EEDF, and Section A.3 lists their Maxwellian values. A row that reads a band prints the value v0_2 runs by default and the band's name, and Section A.4 lists the bands. Each row's full citation is in `v0_2/network.py`. The row ids are the ones the code and the system map use.

A.1 The basic network, 44 rows

8 of them read a band.

row	reaction	rate	*T*	source
G1	e^- + Kr → e^- + Kr^*	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1, the whole 5s manifold at 9.915 eV
C1a	e^- + Kr → e^- + Kr^*	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 5p group at 11.30 eV, prompt cascade to 5s
C1b	e^- + Kr → e^- + Kr^*	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 d+p group at 11.998 eV, as C1a
C1c	e^- + Kr → e^- + Kr^*	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 higher group at 12.75 eV, as C1a
C2	e^- + Kr → 2e^- + Kr^+	$\langle\sigma v\rangle$, Table A.3	T_e	Biagi v7.1 ionization at 13.996 eV
G6	e^- + Kr, elastic	energy loss $(2m/M)\langle\sigma_m v\varepsilon\rangle$	T_e, T_g	Biagi v7.1 momentum transfer
ATT	e^- + F_2 → F^- + F	$\langle\sigma v\rangle$, Table A.3	T_e	SIGLO F_2 set

row	reaction	rate	T	source
DIS_a3	$e^- + F_2 \rightarrow e^- + 2F$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO $a\,^3\Pi_u$, repulsive, so excitation dissociates
DIS_A1	$e^- + F_2 \rightarrow e^- + 2F$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO $A\,^1\Pi_u$, as DIS_a3
VIB1	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 1
VIB2	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 2
VIB3	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 3
VIB4	$e^- + F_2 \rightarrow e^- + F_2$	$\langle \sigma v \rangle$, Table A.3	T_e	SIGLO, v = 4
G2	$2Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^{3}s^{-1}, band kr_pen		Kannari 1985 Table I Penning block, from Leichner and Ericson 1974. Band down to Werner's 1.5×10^{-10}, via Jancaitis 1983
G3	$Kr^* + F_2 \rightarrow KrF(B) + F$	4.73×10^{-10} cm^{3}s^{-1}		rate from Velazco, Kolts and Setser 1976 and Ueda 1984, KrF* yield 0.90 from Velazco, Kolts and Setser 1976, B share from Yu and Setser 1990
G3C	$Kr^* + F_2 \rightarrow KrF(C) + F$	2.9×10^{-10} cm^{3}s^{-1}		as G3, with Yu and Setser's k_B/k_C of 1.6
G3X	$Kr^* + F_2 \rightarrow Kr + 2F$	8.47×10^{-11} cm^{3}s^{-1}		as G3, the KrF*-free rest, assigned to Kr + 2F
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.4×10^{-31} cm^{6}s^{-1}	$(T_g/300)^{-0.75}$	the mean of three measurements, 2.4×10^{-31}. T_g law from Perez-Rios and Greene 2018
R11p	$e^- + Kr_2^+ \rightarrow Kr^* + Kr$	1.6×10^{-6} $(T_e/300\,\mathrm{K})^{-0.55}$ cm^{3}s^{-1}	$T_e^{-0.55}$	Shiu and Biondi 1977, 1.6×10^{-6} $(300\,\mathrm{K}/T_e)^{0.55}$, $\pm$ 10 %. Mikus 1978 within 8 %
K26	$Kr^+ + F^- \rightarrow KrF(B)$	2.95×10^{-6} cm^{3}s^{-1} at 1 atm, 300 K, unscreened	N, screening, $(T_g/300)^{-3}$ on the 3-body part	Kannari 1985, Flannery turnover with Debye screening, and $(T_g/300)^{-3}$ on its three-body part from Thomson theory. Measured with F^- only below 300 Torr, Jacob et al. 1978. No measurement near 1 atm
K108	$Kr_2^+ + F^- \rightarrow KrF(B) + Kr$	2.95×10^{-6} cm^{3}s^{-1} at 1 atm, 300 K, unscreened	N, screening, $(T_g/300)^{-3}$ on the 3-body part	as K26
G4	$KrF(B) \rightarrow Kr + F + h\nu$	1.38×10^{8} s^{-1}		Kannari 1985 Table I, the 96.6 % B-X share of τ_B = 6.99 ns. Eden 1978 measured 6.8 $\pm$ 0.2 ns
G4A	$KrF(B) \rightarrow Kr + F + h\nu$	4.84×10^{6} s^{-1}		$A_{B\text{-}A}/A_{B\text{-}X}$ = 0.035, Dunning and Hay's calculation via Yu and Setser 1990. 272 nm is a theoretical position
G4C	$KrF(C) \rightarrow Kr + F + h\nu$	2.0×10^{7} s^{-1}		ML joint fit. Hakuta 1990 measured 60 $\pm$ 10 ns. Band at 275 nm, printed
BC	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	1.05×10^{-11} cm^{3}s^{-1}		ML joint fit. Hakuta 1990's 1.2×10^{-10} is held out
CB	$KrF(C) + Kr \rightarrow KrF(B) + Kr$	9.55×10^{-12} cm^{3}s^{-1}	$1/K_{eq}(T_g)$	detailed balance of BC, with the joint fit's K_{eq} and Yu and Setser's measured 120 $\pm$ 50 cm^{-1} gap
G5	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	1.9×10^{-30} cm^{6}s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit. T_g law from Shui's calculation for 300-800 K, measured only to 425 K
G5C	$KrF(C) + 2Kr \rightarrow Kr_2F^* + Kr$	3.79×10^{-31} cm^{6}s^{-1}	$(T_g/300)^{-3.32}$	ML joint fit
QE	$KrF(B) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8} cm^{3}s^{-1}		Trainor and Jacob's measured product, re-read with this network's lifetime
QEC	$KrF(C) + e^- \rightarrow Kr + F + e^-$	6.6×10^{-8} cm^{3}s^{-1}		as QE

row	reaction	rate	T	source
QF2	KrF(B) + F_2 → Kr + 3F	4.75×10^{-10} cm^3s^{-1}		ML joint fit. Xu 1993 printed 3.0 ± 0.3×10^{-10} with fast mixing
QF2C	KrF(C) + F_2 → Kr + 3F	3.04×10^{-10} cm^3s^{-1}		ML joint fit
QKR	KrF(B) + Kr → 2Kr + F	1.0×10^{-14} cm^3s^{-1}		ML joint fit, at its lower bound
QKRC	KrF(C) + Kr → 2Kr + F	8.37×10^{-13} cm^3s^{-1}		ML joint fit
C3	Kr_2F^* → 2Kr + F + $h\nu$	4.5×10^6 s^{-1}		Xu, Gadomski and Setser 1993, 220 ± 20 ns. Band at 420 nm, printed
Q2F	Kr_2F^* + F_2 → 2Kr + F + F_2	1.2×10^{-10} cm^3s^{-1}, band kr2f_f2		Xu 1993 Table I, 1.2 ± 0.2×10^{-10}. Band up to Quigley and Hughes' 4.3×10^{-10}
Q2FF	Kr_2F^* + F → 2Kr + 2F	0, band kr2f_f		no Kr measurement, so 0 unless a band run sets it, 0 to 1.8×10^{-10}. The top is Jancaitis 1983's Ar_2F^* analog
FF	2F + Kr → F_2 + Kr	2.8×10^{-34} cm^6s^{-1}		Cohen and Bott 1976 evaluation, 2.8×10^{-34}, every rare gas given the Ar value. Held at its 300 K value
AK8_Kr	Kr^* + 2Kr → Kr_2^* + Kr	1.7×10^{-32} cm^6s^{-1}, band kr2_form	$(T_g/300)^{-0.64}$	Ueda 1984 Table V, 1.7 ± 0.2×10^{-32} at 0.5-1 atm. T_g law from Wieme and Lenaerts 1981, 300-451 K. Band up to Hughes 1977, 5.5×10^{-32}
KR2_RAD	Kr_2^* → 2Kr + $h\nu$	3.7×10^6 s^{-1}		Morikawa 1989, 270 ± 5 ns, and five other measurements
NK7	Kr_2^* + F_2 → KrF(B) + Kr + F	1.05×10^{-10} cm^3s^{-1}, band kr2_f2, half		half of Quigley and Hughes 1978, 2.1 ± 0.2×10^{-10}. The even split is Kannari's. Band up to Chen and Payne's 5.0×10^{-10}
K2F_BR	Kr_2^* + F_2 → Kr_2F^* + F	1.05×10^{-10} cm^3s^{-1}, band kr2_f2, half		the other half of Quigley and Hughes 1978
K2F_F	Kr_2^* + F → KrF(B) + Kr	0, band kr2_f		no measurement, so 0 unless a band run sets it, 0 to 1×10^{-9}. Kannari 1985's 3.0×10^{-10} is an estimate
PEN_KR2	$2Kr_2^*$ → Kr_2^+ + 2Kr + e^-	0, band kr2_pen		no measurement, so 0 unless a band run sets it, 0 to 5.0×10^{-10}. Kannari 1985 took 5.0×10^{-10} from an unpublished 1973 SRI report

A.2 Optional group on the basic network, 2 rows

The group electron is off at the reference point and switches on with `--add`.

row	reaction	rate	T	source
KSTEP	e^- + Kr^* → Kr^+ + $2e^-$	1.56×10^{-7} $T_e^{0.71}\times$ $e^{-4.09/T_e}$ cm^3s^{-1}	T_e	Chen R9 template, an estimate
SUPER	e^- + Kr^* → e^- + Kr	from G1 by detailed balance	T_e	detailed balance of G1, statistical weight ratio 1/12

A.3 Electron-impact rate coefficients

Maxwellian $\langle\sigma v\rangle$ in cm^3s^{-1} at the T_e of each column. The last column is the factor the two-temperature tail applies at the reference point's mid-pulse state, T_e = 3.81 eV with the tail at 0.16 T_e. The rows below the rule give the energy an attached electron takes from the electron gas.

row	process	T_e (eV)					tail
		0.5	1	1.5	2	3	factor
G1	Kr, 5s manifold, 9.915 eV	1.24×10^{-17}	3.44×10^{-13}	1.11×10^{-11}	6.41×10^{-11}	3.78×10^{-10}	0.036
C1a	Kr, 5p group, 11.3 eV	5.1×10^{-19}	5.93×10^{-14}	2.97×10^{-12}	2.11×10^{-11}	1.48×10^{-10}	5.53×10^{-3}
C1b	Kr, d+p group, 11.998 eV	2.5×10^{-19}	5.84×10^{-14}	3.77×10^{-12}	3.08×10^{-11}	2.53×10^{-10}	1.81×10^{-3}
C1c	Kr, higher group, 12.75 eV	9.15×10^{-21}	7.73×10^{-15}	8.8×10^{-13}	1.02×10^{-11}	1.29×10^{-10}	1.48×10^{-4}
C2	Kr, ionization, 13.996 eV	8.06×10^{-21}	1.57×10^{-14}	2.28×10^{-12}	2.94×10^{-11}	4.14×10^{-10}	5.09×10^{-5}
ATT	F_2, attachment	6.58×10^{-9}	3.51×10^{-9}	2.25×10^{-9}	1.61×10^{-9}	9.75×10^{-10}	1
DIS_a3	F_2, $a\,^3\Pi_u$, 3.16 eV	3.01×10^{-12}	1.13×10^{-10}	3.95×10^{-10}	7.37×10^{-10}	1.35×10^{-9}	1
DIS_A1	F_2, $A\,^1\Pi_u$, 4.34 eV	2.06×10^{-13}	1.93×10^{-11}	8.88×10^{-11}	1.9×10^{-10}	4.02×10^{-10}	1
VIB1	F_2, v = 1, 0.111 eV	1.77×10^{-9}	1.22×10^{-9}	8.72×10^{-10}	6.61×10^{-10}	4.27×10^{-10}	1
VIB2	F_2, v = 2, 0.219 eV	2.01×10^{-9}	1.49×10^{-9}	1.09×10^{-9}	8.37×10^{-10}	5.47×10^{-10}	1
VIB3	F_2, v = 3, 0.324 eV	1.21×10^{-9}	9.75×10^{-10}	7.35×10^{-10}	5.71×10^{-10}	3.78×10^{-10}	1
VIB4	F_2, v = 4, 0.421 eV	3.83×10^{-10}	3.27×10^{-10}	2.51×10^{-10}	1.96×10^{-10}	1.31×10^{-10}	1
ATT	energy an attached e^- takes, eV	0.37	0.46	0.53	0.59	0.72	
	$\frac{3}{2}T_e$, for comparison	0.75	1.50	2.25	3.00	4.50	

A.4 The bands

`network.BANDS`. `Params.rates` sets a band, and `Params.trimer_tg_pow` the trimer law.

band	process	default	range	source
kr2_f	Kr_2^* + F → KrF(B) + Kr	0	0 to 1.0×10^{-9} cm^3s^{-1}	No measurement. Kannari's 3.0×10^{-10} traces to SRI MP-76-99, whose model leaves the reaction out (p.45) and whose one number is a conditional 3-10×10^{-10} for Ar_2^* + F (p.65), the band's top. Jancaitis 1983 has no such row
trimer_tg_pow	the trimer's $(T_g/300)^n$	-3.32	-3.32 to -0.50 exponent	Shui's calculation over 300-800 K, the default, measured to 425 K. Salesky and Kimura inferred -0.5 from pure-Kr efficiencies, a bound that never sets it. Params.trimer_tg_pow sets it
kr_pen	Kr^* + Kr^* → Kr^+ + Kr + e^-	5.0×10^{-10}	1.5×10^{-10} to 5.0×10^{-10} cm^3s^{-1}	Kannari 1985 gives 5.0×10^{-10} from Leichner and Ericson 1974, v0's and the default. Jancaitis's model carries 1.5×10^{-10} from Werner's thesis (1975)
kr2f_f2	Kr_2F^* + F_2	1.2×10^{-10}	1.2×10^{-10} to 4.3×10^{-10} cm^3s^{-1}	Xu 1993 measured $1.2 \pm 0.2\times10^{-10}$, the default, and lists 1.5, 1.8 and 4.3×10^{-10} before it. Jancaitis 1983 measured $1.8 \pm 0.2\times10^{-10}$
kr2f_f	Kr_2F^* + F → 2Kr + 2F	0	0 to 1.8×10^{-10} cm^3s^{-1}	No Kr measurement. Jancaitis 1983 found F atoms quench Ar_2F^* at 0.5-1 times F_2 (p.153). With his Kr_2F^* + F_2, 1.8×10^{-10}, that analog gives 0.9-1.8×10^{-10} and sets the top
kr2_form	Kr^* + 2Kr → Kr_2^* + Kr, 300 K	1.7×10^{-32}	1.7×10^{-32} to 5.5×10^{-32} cm^6s^{-1}	Ueda 1984 measured $1.7 \pm 0.2\times10^{-32}$ at 0.5-1 atm, the default. Low-pressure afterglows give 1.79 to 4.4×10^{-32} and Hughes 1977 5.5×10^{-32}
kr2_f2	Kr_2^* + F_2, both channels	2.1×10^{-10}	2.1×10^{-10} to 5.0×10^{-10} cm^3s^{-1}	Quigley and Hughes 1978 measured $2.1 \pm 0.2\times10^{-10}$, the default. SRI MP-76-99 gives 3.0×10^{-10} and Chen and Payne 1979 5.0×10^{-10}, beyond each other's errors

band	process	default	range	source
kr2_pen	$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0	0 to 5.0×10^{-10} cm^3s^{-1}	No measurement. Kannari's 5.0×10^{-10} is Lorents and Olson's unpublished 1973 SRI report, which gives Ar_2^* the same value, and Jancaitis's model carries 1.0×10^{-10} from Werner et al. 1977

A.5 Where v0_2's v0 network differs

The v0 network keeps the notes' six reactions with REBUILD's values, and its optional groups, quench, kr2, ions, electron, bc, f2e, are REBUILD's. These rows differ from v0_2's basic network or exist only in the v0 network. The rows ending in _C are the bc group's copies for KrF(C).

row	reaction	rate	T	source
G2	$2Kr^* \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I, Penning block
G3	$Kr^* + F_2 \rightarrow KrF(B) + F$	8.1×10^{-10} cm^3s^{-1}		Kannari 1985 Table I, all to B. The report it credits gives 7.2×10^{-10}
K15	$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.5×10^{-31} cm^6s^{-1}	$(T_g/300)^{-0.75}$	Kannari 1985, Boichenko 2000
R11p	$e^- + Kr_2^+ \rightarrow Kr^* + Kr$	2.24×10^{-7} $T_e^{-1/2}$ cm^3s^{-1}	T_e	Kannari 1985 Table I, 1.39×10^{-6} at 300 K on a $T_e^{-1/2}$ template
K26	$Kr^+ + F^- \rightarrow KrF(B)$	2.95×10^{-6} cm^3s^{-1} at 1 atm, 300 K, unscreened	N, screening	Kannari 1985, Flannery turnover with Debye screening, no T_g law
K108	$Kr_2^+ + F^- \rightarrow KrF(B) + Kr$	2.95×10^{-6} cm^3s^{-1} at 1 atm, 300 K, unscreened	N, screening	as K26
G4	$KrF(B) \rightarrow Kr + F + h\nu$	1.43×10^{8} s^{-1}		Kannari 1985 Table I, all of τ_B as B-X
G5	$KrF(B) + 2Kr \rightarrow Kr_2F^* + Kr$	6.0×10^{-31} cm^6s^{-1}	$(T_g/300)^{-3.32}$	Xu, Gadomski and Setser 1993 Table II, $6.0 \pm 2.0\times10^{-31}$
G5_C	$KrF(C) + 2Kr \rightarrow Kr_2F^* + Kr$	6.0×10^{-31} cm^6s^{-1}	$(T_g/300)^{-3.32}$	as G5, applied to KrF(C)
AK8_Kr	$Kr^* + 2Kr \rightarrow Kr_2^* + Kr$	2.7×10^{-32} cm^6s^{-1}		REBUILD's estimate, Kannari's Ar-partner rate given to Kr
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$	8.0×10^{7} s^{-1}		Kannari 1985 Table I, 8.0×10^{7} per second
NK7	$Kr_2^* + F_2 \rightarrow KrF(B) + Kr + F$	3.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I
K2F_BR	$Kr_2^* + F_2 \rightarrow Kr_2F^* + F$	3.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I
K2F_F	$Kr_2^* + F \rightarrow KrF(B) + Kr$	3.0×10^{-10} cm^3s^{-1}		Kannari 1985, an estimate. No measurement exists
PEN_KR2	$2Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	5.0×10^{-10} cm^3s^{-1}		Kannari 1985 Table I, from an unpublished 1973 SRI report that gives Ar_2^* the same value. No measurement held
K33	$KrF(B) + Kr \rightarrow 2Kr + F$	4.0×10^{-12} cm^3s^{-1}		Xu, Gadomski and Setser 1993, $4.0 \pm 0.5\times10^{-12}$
K33_C	$KrF(C) + Kr \rightarrow 2Kr + F$	4.0×10^{-12} cm^3s^{-1}		as K33, applied to KrF(C)
K32	$KrF(B) + F_2 \rightarrow Kr + 3F$	3.0×10^{-10} cm^3s^{-1}		Xu, Gadomski and Setser 1993, $3.0 \pm 0.3\times10^{-10}$
K32_C	$KrF(C) + F_2 \rightarrow Kr + 3F$	3.0×10^{-10} cm^3s^{-1}		as K32, applied to KrF(C)
K67	$KrF(B) + e^- \rightarrow Kr + F + e^-$	4.5×10^{-8} cm^3s^{-1}		Kushner, Hanson and Schneider 1989
K67_C	$KrF(C) + e^- \rightarrow Kr + F + e^-$	4.5×10^{-8} cm^3s^{-1}		as K67, applied to KrF(C)
K125	$Kr_2F^* + F_2 \rightarrow 2Kr + F + F_2$	1.5×10^{-10} cm^3s^{-1}		Kannari 1985 Table I
K_2Fe	$Kr_2F^* + e^- \rightarrow 2Kr + F + e^-$	1.0×10^{-7} cm^3s^{-1}		Kannari 1985, estimated there
BC_Kr	$KrF(B) + Kr \rightarrow KrF(C) + Kr$	5.0×10^{-10} cm^3s^{-1}		Kannari's B-C mixing with a Kr partner, an estimate
CB_Kr	$KrF(C) + Kr \rightarrow KrF(B) + Kr$	3.41×10^{-10} cm^3s^{-1}		detailed balance of BC_Kr at 300 K

row	reaction	rate	T	source
URAD_C	KrF(C) $\rightarrow$ Kr + F + $h\nu$	1.33×10^{7} s^{-1}		Kannari 1985 Table I, 1.33×10^{7} per second

A.6 Formation energies

Energies above ground-state Kr and F_2 at rest. Every reaction's energy release is the difference of these numbers, so no row can make or lose energy. c_v is the heat capacity per particle in units of k_B.

species	E_{form} (eV)	c_v	what sets it
Kr	0	1.5	ground state
F_2	0	2.5	ground state
F	0.8	1.5	half the F_2 bond, D_0 = 1.60 eV
Kr*	9.915	1.5	Biagi's 5s threshold, the 5s[3/2]2 level
Kr$^+$	13.996	1.5	Biagi's ionization threshold
KrF(B)	5.856	2.5	$D_e(B)$ = 5.54 eV below the ion pair, Tellinghuisen 1976
KrF(C)	5.84112	2.5	120 cm^{-1} below B, Yu and Setser 1990
Kr_2F^*	5.276	3	0.58 eV below KrF* + Kr, calculated by Huestis et al. 1984
Kr_2^*	9.2	2.5	the 147-nm excimer, an estimated well
Kr_2^+	12.846	2.5	IP(Kr) less $D_0(Kr_2^+)$ = 1.15 eV, measured by photoionization, Ng et al. 1977
F^-	-2.6	1.5	E_{form}(F) less the electron affinity, 3.40 eV